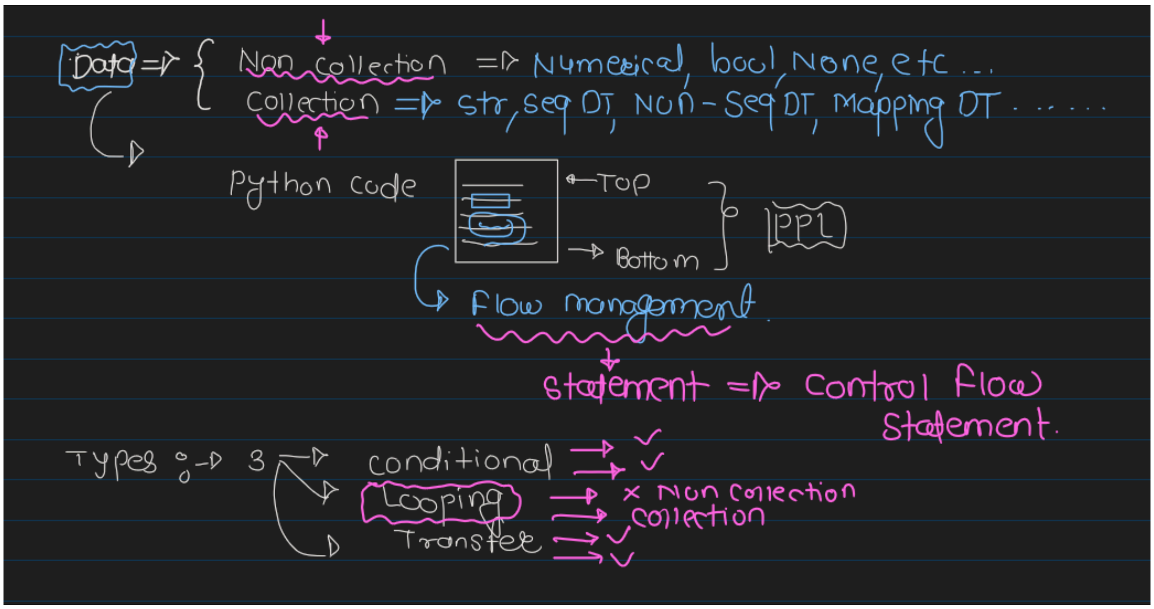
# Control Flow Statemet :

```
A Control flow Statement in Python is a Block of code that
determines the order in which Statements are executed based on
certain conditions or loop.
Thses statement allow you to control the flow of execution within
our program. enabling you to make decision , repeat action and
handle executional situation.
```




**There are Three Main Types of Control flow Statement :**

```
1. Conditional Statement :
    1.1 if
    1.2 elif
    1.3 else
2. Looping Control Flow Statement :
    2.1 For
    2.2 While
3. Transfer Control Flow Statement :
    3.1 pass
    3.2 Continue
    3.3 Break
```

In [13]:

```
number=range(1,40)
for i in number :
    # Transfer statement : Break  # loop end
    if i == 5 or i== 10 :
        print('Encountered 5 , 10 --> Loop stopped')
        continue  # skip
        # pass     ---> its just a place holder
    print(i)
```

```
1
2
3
4
Encountered 5 , 10 --> Loop stopped
6
7
8
9
Encountered 5 , 10 --> Loop stopped
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
```

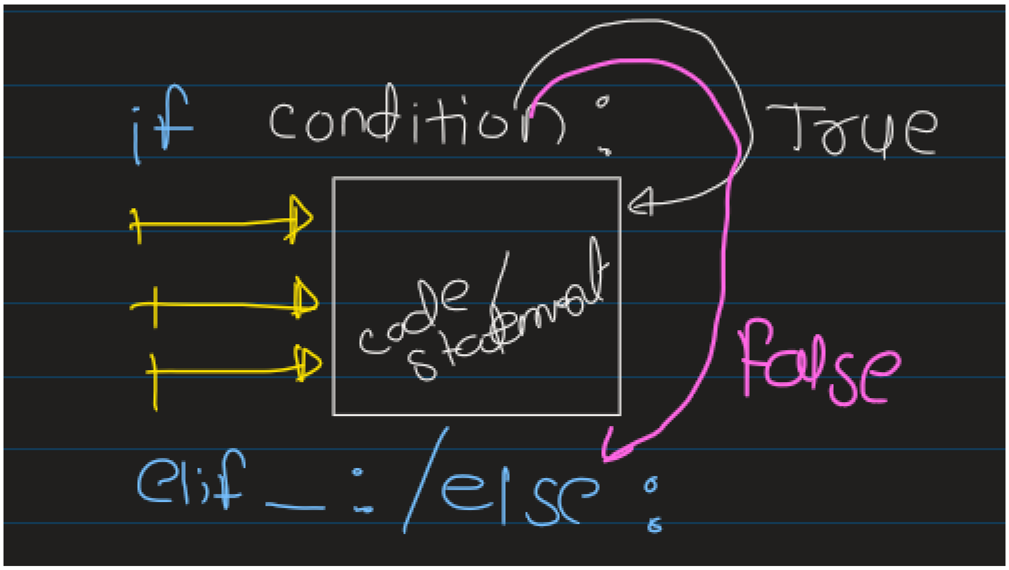
# 1. Conditional Control Flow Statement :

```
On Particular condition
it is going to decide wheter next statement need to execute or not.
IF condition statified will  execute the if block  or othewise
another block {elif or else}.
    1.1 if statement
    1.2 elif statement
    1.3 else statement
```

### 1.1 if statement :

```
        It is used to decide wheter a certain statement or block of
statement will be execute or not.

    if condition is True will execute if block of code / statement
.
    if condition is False then will execute another block of code /
statement .
Syntax :
```




In [16]:
```python
# WAP To check the user input value is even
value=int(input('Please enter value ='))
if value%2==0:
    print('This is Even number',value)
```

In [20]:
```python
# WAP to chekc the user input value is divided by 3 and  5 .
value=int(input('Pls enter value ='))
if value % 3==0 and value%5==0 :
    print(f'{value} is divided by 3 and 5')
```

```
30 is divided by 3 and 5
```

## 1.2 if_else:

```
if condition is True then will be execute .
if not True will execute else block.

Syntax :
    if Condition:
        if_statement
    else:
        else_statement
```

In [21]:
```python
# WAP to chekc the user input value is divided by 3 and  5 .
value=int(input('Pls enter value ='))
if value % 3==0 and value%5==0 :
```

```python
    print(f'{value} is divided by 3 and 5')
else:
    print(f'{value} is not divided  by 3 and 5')
```

```
10 is not divided  by 3 and 5
```

## 1.3 if __elif___else :

```
When we have Multiple Conditions to check then  we used elfi block
and any one statement will be execute from if elif else block
```

In [25]:

```python
int_obj=None
float_obj=None
str_obj=None
value=eval(input('pls enter a obj = '))
if type(value) == int :
    int_obj=value
    print(f'This is an Int Vlaue {int_obj}')
elif type(value) == float :
    float_obj=value
    print(f'This is a float Vlaue {float_obj}')
elif type(value) == str :
    str_obj=value
    print(f'This is a str Vlaue {str_obj}')
else:
    print('This is Another obj',type(value))
```

```
This is an Int Vlaue 10
```

# Q1. WAP to accept two diff numbers from user and print the largest number among this . # Q2. Traffic Light simulator WAP That takes a color as input('red','Yellow','Green') and print the corresponding action : if red ----> print('Stop') if yellow --> Get Ready if green ---> Go -----> for any other input print --> Invalid color # Q3. Age wise category find WAP that takes a persion age as input and print their category based on

# Nested if_else :

```
it is an if or else statement placed inside another if or else
block it is when we need to check multiple conditions step by step
before making a final decision.
```

###### syntax : if condition1: if condition2: # code if both conditions are true else: # code if condition1 is True but condition2 is False else: # code if condition1 is False

# WAP Student eligibility

```
std age is more than 18 and marks is more than and equal to 60 then
procceed this student  for admmision.
```

In [3]:

```python
age= int(input('Please enter student age = '))
marks= float(input('Enter std marks = '))
if age >18 :
    if marks>=60:
        print('Eligible for Admission')
```

```
    else :
        print('Marks are less')
else:
    print('Age is less than 18')
```

```
Marks are less
```

# Question: Engineering and Medical Admission with Branch Selection (Nested if-else)

```
Write a Python program to check a student’s admission. First,
accept the student’s overall percentage. If the percentage is 60%
or above, ask the student to choose Engineering or Medical.
        •       If the student selects Engineering, ask them to
choose one of the following branches:
        •       Computer Engineering (Minimum PCM: 85%)
        •       Mechanical Engineering (Minimum PCM: 75%)
        •       Civil Engineering (Minimum PCM: 70%)
        •       If the student selects Medical, ask them to choose
one of the following branches:
        •       MBBS (Minimum PCB: 90%)
        •       BDS (Minimum PCB: 85%)
        •       BAMS (Minimum PCB: 75%)
```

In [15]:

```
Marks= float(input('Enter The 12the pct marks = '))
if Marks >=60 :
    stream=input("""
        Choose stream
            1. Engineering
            2. Medical
    """)
    ######################################################################
    if stream.lower() == 'engineering':
        print('Congo! ur elibigible for engineering admission')
        branch=input("""
        ------------- Choose any One Branch -------------
            1.Computer
            2. Mechanical
            3. Civil
        """)
        pcm = float(input('Enter PCM pct = '))
        if branch.lower() == 'computer':
            if pcm>85:
                print('Admission Confirmed in COMP engineering')
            else:
                print('Not Eligible for COMP Engineering')
        elif branch.lower() == 'mechanical':
            if pcm >=75:
                print('Admission Confirmed in Mechanical engineering')
            else:
                print('Not Eligible for mechanical Engineering')
```

```python
        elif branch.lower() == 'civil':
            if pcm>70:
                print('Admission Confirmed in civil engineering')
            else:
                print('Not Eligible for civil Engineering')
        else:
            print('Invalid engineering branch')


    elif stream.lower() == 'medical':
        print('Congo! ur elibigible for medical admission')
        branch= input(
            """
            ------------- Choose Branch -------------
                1. MBBS
                2. BDS
                3. BAMS
            """
        )
        pcb = float(input('Enter PCB pct = '))
        if branch.lower() == 'mbbs':
            if pcb >=90:
                print('Admission confimed in MBBS')
            else:
                print('NOT confiremed for MBBS')
        elif branch.lower():
            if pcb >= 85:
                print('Admission confimed in BDS')
            else:
                print('NOT confiremed for BDS')
        elif branch.lower() == 'bams':
            if pcb >= 75:
                print('Confiremed for BAMS')
            else:
                print('Not confirmend for BAMS')
        else:
            print('Invalid Medical Branch')
    else:
        print('Invalid Stream')

else :
    print('Not Eligible for Admission (bcz Marks must be at least 60)')
```

```
Congo! ur elibigible for medical admission
Admission confimed in MBBS
```

# Looping control flow statemet :

```
Looping Control flow is a Programming concept usd to execute a
block of code repeatedly until a specified condition becomes false
or far a fixed number of iterations.

in Python loops helps us automate repetitive tasks instead of
writing the same code multiple times.
```

```
## Types of Loops In Python :
    1. for Loop  :
        A for loop is used when the Number of iteration  is known
                for Iterative_variable in iterable_object:
                            ### code
    2. while loop :
        A while loop execute as long as the given condition is True
.
                while condition :
                    ## code
```

In [19]:
```
'Python'
```

```
1
0
```

In [23]:
```
print('python')
print('python')
print('python')
print('python')
print('python')
```

```
python
python
python
python
python
```

In [1]:
```
import time
for i in range(1,6,1):  # 1,2,3,4,5
    print(i,'Python')
    time.sleep(2)
```

```
1 Python
2 Python
3 Python
4 Python
5 Python
```

for ---> Keyword that strat the loop i ---> iterative_variable stores one element from the collection obj in each iterable. in ----> its membership operatior --> connects the variable to the seq of element .### flow of execution start | Take first value | execute the code | Take next value | execute the code | continue | seq finished | stop

In [2]:
```
for i in range(10,0,-1):  # 1,2,3,4,5
    print(i,'Python')
```

```
10 Python
9 Python
8 Python
7 Python
6 Python
5 Python
4 Python
3 Python
2 Python
1 Python
```

In [5]:
```python
# WAP print the even value square and print the odd value cube --> range(1,40)
for value in range(1,41):
    # even value print and then square of this value
    if value%2==0 :
        print(f'{value} of square is {value**2}')
    else:
        print(f'{value} of cube is {value**3}')
```

```
1 of cube is 1
2 of square is 4
3 of cube is 27
4 of square is 16
5 of cube is 125
6 of square is 36
7 of cube is 343
8 of square is 64
9 of cube is 729
10 of square is 100
11 of cube is 1331
12 of square is 144
13 of cube is 2197
14 of square is 196
15 of cube is 3375
16 of square is 256
17 of cube is 4913
18 of square is 324
19 of cube is 6859
20 of square is 400
21 of cube is 9261
22 of square is 484
23 of cube is 12167
24 of square is 576
25 of cube is 15625
26 of square is 676
27 of cube is 19683
28 of square is 784
29 of cube is 24389
30 of square is 900
31 of cube is 29791
32 of square is 1024
33 of cube is 35937
34 of square is 1156
35 of cube is 42875
36 of square is 1296
37 of cube is 50653
38 of square is 1444
39 of cube is 59319
40 of square is 1600
```

In [8]:
```python
name='Shivsharan'
count=0
for i in name :
    count+=1
    print(count,i)
print(count)
```

```
1 S
2 h
3 i
4 v
5 s
6 h
7 a
8 r
9 a
10 n
10
```

In [2]:
```
len(name)
```

Out[2]:
```
10
```

# Print the star pattern * ** *** **** *****

In [10]:
```
for i in range(1,6):
    print(i*'*')
```

```
*
**
***
****
*****
```

***** ***** ***** ***** *****

In [12]:
```
for i in range(1,6):
    print(5*'*')
```

```
*****
*****
*****
*****
*****
```

In [38]:
```
for i in range(1,6):  # row ---> 5 --> 1,2,3,4,5
    for j in  range(1,6):  # column 5  --> 1,2,3,4,5
        print('*',end=' ')
    print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

# While loop :

```
A while loop repeatedly execute a block of code as longh as the
given  condition evaluate to True.

Syntax :
    while  condition :
        Statement
```

1 2 3 4 5

In [2]:
```
i=1
while i<=5: # True ,True ,True,True,True,FaLSE
    print(i) # 1,2,3,4,5
    i=i+1  # 2,3,4,5,6
```

```
1
2
3
4
5
```

In [3]:
```
count=5
while count>0:
    print(count)
    count-=1
```

```
5
4
3
2
1
```

In [7]:
```
# WAP Print the Even Number and odd number from 1 to 20
```

In [6]:
```
i=1
while i<=20:
    if i%2==0:
        print(f'even number ={i}')
    else:
        print(f'odd number = {i} ')
    i+=1
```

```
odd number = 1
even number =2
odd number = 3
even number =4
odd number = 5
even number =6
odd number = 7
even number =8
odd number = 9
even number =10
odd number = 11
even number =12
odd number = 13
even number =14
odd number = 15
even number =16
odd number = 17
even number =18
odd number = 19
even number =20
```

In [10]:
```
# Factorial of a Number
n=int(input('Enter Number = '))
```

```
fact=1
while n>0:
    fact*=n
    n-=1

print('Factorial = ',fact)
```

```
Factorial =  1
```

# 3. Jumping Control Flow Statement :

```
Jumping control flow statements are special statements in Python
That change the Normal execution of a loop or program . They allow
the program to skip , stop or do nothing when certain conditions
are met.
Python provides Three main jumping statement .
3.1 pass
3.2 continue
3.3 break
```

# 3.1 pass statemnt :

```
The Pass Statement is a Null statement . It does nothing when
execute. it is used as a placeholder where python expects a
statement syntactically , But we dont want to implement the code
yes.
```

In [15]:
```
if 12>23:
    pass
```

In [19]:
```
for i in range(7):
    if i==3:
        pass
    else:
        print(i)
```

```
0
1
2
4
5
6
```

# 3.2 Continue Statement :

```
The Continue statement skips the current iteration of the loop and
Immediately moves to the next iteration. The Loop itself does not
stop .
```

In [3]:
```
i=1
while i<=10:
    if i==5:
        i+=1
        continue
    print(i)
    i+=1
```

```
1
2
3
4
6
7
8
9
10
```

In [4]:
```
for i in range(1,11):
    if i==5:
        continue
    print(i)
```

```
1
2
3
4
6
7
8
9
10
```

# 3.3 break :

    The break Statement is used to immediately terminate(stop) a loop
    (for or while).when python encounters break ,it exits the loop and
    execute the first statement after the loop.

In [5]:
```
for i in range(1,11):
    if i==5:
        break
    print(i)
```

```
1
2
3
4
```

In [6]:
```
# Login system :
password='Python'
while True:
    user = input('Enter password : ')
```

```
    if user== password:
        print('Login Sucessful')
        break
    print('Wrong Password')
```

```
Wrong Password
Login Sucessful
```

# WAP ATM machine Simulation

```
Use control flow :
    while loop , if elif and else and break

Write a Python Program  to simulate an ATM   sys with the following
menu.
    - Check Balance
    - Deposite Money
    - Withraw money
    - exit
```

In [2]:

```
Balance=10000
while True :
    print('           ATM Menu           ')
    print('1. Check the Balance')
    print('2. Deposit Money')
    print('3. Withraw Money ')
    print('4. Exit')

    choice= int(input('enter Your choice'))
    if choice ==1 :
        print(f'Balance = {Balance}')
    elif choice==2:
        amount=float(input('Enter Deposit Amount'))
        if amount>0:
            Balance= Balance+amount
            print(f'{amount} , Deposited Sucessfuly')
            print(f'Updated Balance {Balance}')
        else:
            print('Invalid Amount')
    elif choice==3 :
        amount=float(input('Enter withraw amount'))
        if amount <=Balance:
            Balance=Balance-amount
            print(f'{amount} withrawn ')
            print('Remaning Balance = ',Balance)
        else:
            print('Insufiicient Balance')

    elif choice==4:
        print('Thankq For Using Our ATM card')
        break
    else:
        print('Invalid Choice! PleaSe try Again')
```

```
          ATM Menu
1. Check the Balance
2. Deposit Money
3. Withraw Money
4. Exit
Balance = 10000
          ATM Menu
1. Check the Balance
2. Deposit Money
3. Withraw Money
4. Exit
1200.0 , Deposited Sucessfuly
Updated Balance 11200.0
          ATM Menu
1. Check the Balance
2. Deposit Money
3. Withraw Money
4. Exit
2000.0 withrawn
Remaning Balance =  9200.0
          ATM Menu
1. Check the Balance
2. Deposit Money
3. Withraw Money
4. Exit
Thankq For Using Our ATM card
```

# WAP Student Result Management

```
Control flow statement :
    while , if-elif and else
Enter marks for Multiple student and displayr:
    pass (Marks>=40)
    fails (marks <40)
    stop entering data when  the user type -1
```

# Login sys with maximum 3 Attempts :

```
create a login sys where the suer has only 3 attemopts to enter the
correct unsrname and password . if successful, display login
sucessful otherwise lock the account
```

# Electricity Bill Calculaot :

```
COntrol Flow :
if_elfi_else
calculate the bill :
1st 100 unit --> 5rs /unit
```

next 100 unit --> 7rs / Unit
above 200 unit --> 10 rs /unit